

popquiz impulse-momentum theorem

⚠ This is a preview of the published version of the quiz

Started: Apr 29 at 7:58pm

Quiz Instructions



Question 1 12.5 pts

When a boxer moves into an oncoming punch, the force experienced is
(decrease time of impact)

☐

all of the above

☐

decreased.

☐

increased.

☐

no different, but the timing is different.



Question 2 12.5 pts

Whether a truck comes to a stop by crashing into a haystack or a brick wall, the impulse is
hint: impulse is force x time. is the force the same?

☐

greater with the brick wall.

☐

both the same

☐

greater with the haystack.



Question 3 12.5 pts

When a boxer is moving away from a punch, the force experienced is reduced because

☐

the time of contact is increased.

☐

momentum transfer is reduced.

☐

all of the above



the force is less effective.



Question 4 12.5 pts

The change in momentum that occurs when a 1.0 kg ball traveling at 4.0 m/s strikes a wall and bounces back at 2.0 m/s is



2 kg m/s.



8 kg m/s.



6 kg m/s.



4 kg m/s.



Question 5 12.5 pts

A cannon recoils while firing a cannonball. The speed of the cannon's recoil is relatively small because the

hint: remember baloo and mowgli.



cannon has much more mass than the cannonball.



momentum is mainly concentrated in the cannonball.



force against the cannon is smaller than against the ball.



momentum of the cannon is smaller.



Question 6 12.5 pts

The average braking force of a 1000-kg car moving at 10 m/s braking to a stop in 5 s is

hint: $F t = m \Delta(V)$



5000 N.



3000 N.



4000 N.

☐

1000 N.

☐

2000 N.



Question 7 12.5 pts

The greater efficiency achieved with the curved blades of a Pelton wheel is due to

☐

bouncing.

☐

greater time of water impact.

☐

rotational momentum.

☐

divided impulse.



Question 8 12.5 pts

The force that accelerates a rocket in outer space is exerted on the rocket by the

☐

rocket's nose cone.

☐

rocket's wings.

☐

atmospheric pressure.

☐

none of the above

☐

exhaust gases.

Not saved

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